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Deep Reinforcement Learning-Based Access Control for Buffer-Aided Relaying Systems With Energy Harvesting

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ABSTRACT This paper considered a buffer-aided relaying system with multiple source-destination pairs and a relay node (RN) with energy harvesting (EH) capability. The RN harvests energy from the ambient environment and uses the harvested energy to forward the sources' information packet to the corresponding destinations. It is assumed that information on the EH and channel gain processes is unavailable. Thus, the model free deep reinforcement learning (DRL) method, specifically the deep Q-learning, is applied to learn an optimal link selection policy directly from historical experience to maximize the system utility. In addition, by taking advantage of the structural features of the considered system, a pretraining scheme is proposed to accelerate the training convergence of the deep Q-network. Experiment results show that the proposed pretraining method can significantly reduce the training time required. Moreover, the performance of the transmission policy obtained by using deep Q-learning is compared with that of several conventional transmission schemes. It is shown that the transmission policy obtained by using our proposed model can achieve better performance.

INDEX TERMS Energy harvesting, buffer-aided relaying, Markov decision process, deep reinforcement learning.

I. INTRODUCTION

Cooperative relaying communication, in which a relay node helps to forward the source's information to destination, is capable of attaining significant throughput and reliability improvements [1]. It has also been shown that by adding a data buffer at the relay node (buffer-aided relaying), the throughput and reliability of cooperative relaying system can be further improved [2]–[4]. Usually, the nodes of buffer-aided relaying systems are powered by using power line (e.g., a base station) or battery with limited operation life. However, in many environments, e.g., places which is hard-to-reach, it is costly or even impossible to place dedicated power lines or replace battery regularly. To overcome these challenges, energy harvesting (EH) has been considered as one of the promising technologies [5]. By collecting ambient energy from environments, e.g., solar, wind, thermal, vibration, radio frequency signal, and converting them into electrical energy, EH technology can reduce the dependence on the conventional sources of energy and prolong the lifetimes of EH devices [6].

Hence, EH based buffer-aided relaying has been studied for different network models [7]–[10]. Generally, implementing efficient EH communication should consider the challenges of limited amount of energy that can be harvested and the changing harvestable energy and channel gain with time. In [7], by taking both the quality of the links and the energy stored in the RNs into consideration, the benefits of relay selection on multiple EH based buffer-aided relay nodes (RNs) was investigated. In [8], the achievable throughput of a buffer-aided two-way relaying system with EH were derived. The optimal transmission strategy for a three nodes

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buffer-aided relaying system was studied in [9], where both the source node and the RN harvest energy from the environment. In [10], the delay constraint for a buffer-aided relaying system with energy harvesting was considered and the system throughput was investigated. In these works, it was assumed that the EH and channel gain processes are either known ahead of time (no-causal case) or their distributions are available. However, in practical systems, these information on EH and channel gain processes are usually unknown.

To overcome such challenges, model free technologies has been discussed for designing efficient transmission strategies [11]–[14]. One of the promising approaches used is reinforcement learning (RL), which is known as algorithms enabling to optimize system performance in unknown environments [15], [16]. In [13], an EH point-to-point communications system is investigated. The EH and channel gain processes are modeled as Markov processes and Q-learning was used to learn a transmission power allocation policy that maximizes the amount of data arriving at the destination. The authors of [14] also considered an EH point-to-point communications system in which the energy and data arrivals are formulated as Markov processes. Q-learning is used to find the optimal transmission policy that maximizes the expected total transmitted data when the transmitter is active. In [17], the energy efficient resource allocation strategy was investigated for a body area network by using RL method. In [18] and [19], the actor-critic RL method were adopted to design transmission schemes for wireless communication systems. Specifically, in [18], user scheduling and resource allocation in heterogeneous networks powered by hybrid energy is studied, and actor-critic RL was used to maximize energy efficiency of the overall network. In [19], the problem of energy management for EH wireless sensor nodes is considered, and formulated as an RL problem with continuous state and action spaces. In [20] and [21], distributed RL based communication were investigated for EH networks. All these works have shown the potentiality of applying RL in designing communication schemes for EH communication networks.

In this paper, we consider a buffer-aided relaying system, in which multiple source-destination pairs communicate through the help of a RN. The RN has no reliable power supply and harvests energy from the environment and stores it in its battery to support wireless communication. This system can be treated as a communication system which is temporarily set up in an emergency or hard-to-reach environment. The RN can be a base station which cannot be powered with a power line. It is also assumed that the RN has a data buffer and thus can dynamically choose one link from the source-relay and the relay-destination links in each time slot to deliver data packets. We study how to effectively control the access of each link under the energy harvesting constraint to maximize the system utility. For this system setup, the decision (selection which link to communicate) in current time slot will have impact on the future operations. Thus, it is usually modeled as a Markov decision process (MDP) if its probabilities can be accurately got. However, as we assume no information on the EH and channel gain processes, the model free RL is applied to find the optimal link selection strategy. Since the considered system has large state space, the applying of Q-learning faces the challenge of storing a large Q-value table. Thus, we deploy Deep Q-Learning with pretraining to learn the optimal link access policy directly from historical experience without any prior information about the system dynamics. Deep Q-Learning is a combination of deep learning and reinforcement learning, which is firstly proposed in [22], [23]. The method originally aims to deal with control policy learning with high-dimensional input, such as raw images. The model and its variants have been successfully used in many dynamic decision making domains, especially those with very large state spaces, such as game playing. Since proposed in 2013, Deep Q-Learning has gained much attention, and many variants have been proposed, including double DQN [24], dueling DQN [25], DRQN [26], prioritized DQN [27], bootstrapped DQN [28], etc. Some applications can be found in [29]–[32]. These models differ in network structures, experience replay, ε-greedy or reward function but have the same core infrastructure. The main idea comes from the basic version of Q-learning [33].

The main contribution of this work is as follows:

1) We formulate the link selection problem in the EH based buffer-aided relaying systems as a RL problem. To resolve the optimization problem with Deep Q-learning, we propose to rewrite the problem into a state, action, and reward form.
2) Based on some structured properties of considered system, we proposed a pretraining method to accelerate the training convergence of the proposed Deep Q-learning method. Experiment results show that the proposed pretraining method can significantly reduce the training time.
3) Simulations are conducted to show the performance of the proposed Deep Q-learning method. Experiment results show that as compared with the traditional methods, our model can achieve better performance.

II. SYSTEM MODEL

Figure 1 shows the framework of the system. Suppose that in the system there are N source-destination pairs,

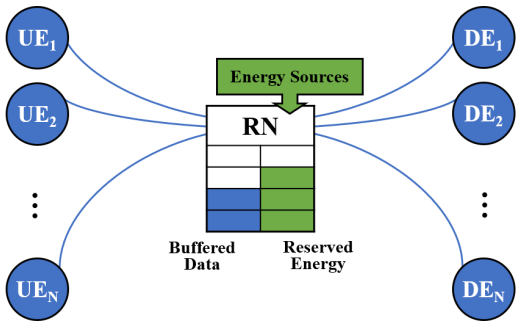


FIGURE 1. A buffer-aided relaying system with energy harvesting.

(UE_i, DE_i) for $i = 1..N$, where RN receives packets from the source nodes via the source-relay links, and forwards these packets to the corresponding destination nodes via the relay-destination links. The transmission rate of each link is discretized into different levels, represented by the amount of packets delivered in each time slot. The RN has a data buffer and a battery, for data and power storage respectively. A certain share of data buffer is allocated for each source-destination pair. The share of the buffer stores those packets that are received from the source node and have not yet forwarded. The energy harvested from the environment is stored in the battery.

At any time t, the relaying system needs to decide which link should be selected for transmission. If a source-relay link is selected, RN receives data packets from the corresponding source node via the link. The amount of received data packets is under the constraints of the quality of the link (transmission rate) and the buffer quota allocated to the source-destination pair. If a relay-destination link is selected, RN forwards the buffered data packets to the corresponding destination node via the link. The amount of forwarded data packets is under the constraints of the quality of the link, existing buffered packets, and the remaining power in the battery. We assume that receiving a packet does not consume energy of the RN and the RN uses fixed transmitting power to forward data packets to the destinations.

The RN has EH capacity and harvests energy from the ambient environment in each time slot t. A battery with certain capacity stores the energy that is harvested from the environment by RN. If the remaining energy in the battery is insufficient for forwarding data, the RN can choose to suspend the transmission and wait for more energy. In this paper, we regard the energy harvest at the RN as a Poisson process with capacity constraint, similar as many previous works [7]–[10], [34], [35].

The relaying system with the data buffer and the battery is capable to memorize historical states and decisions. The system state s^t at a certain time t is completely decided by the previous state s^{t-1} and the action taken at time $t-1$. Hence the task of link selection for the relaying system can be formulated as a Markov decision process.

III. MDP FORMULATION

As stated above, the access control of a buffer-aided relaying system with energy harvest is a Markov decision process. The system decides an action a of access control given current state s, and transitions from s to its successor state s' after taking out the action. In this section, we introduce the MDP formulation of the above relaying system, including the state space, the action space, the transition function and the reward function.

A. STATE SPACE

Let $S = \{s|s = (U, D, B, E)\}$ be the state space, in which a system state s^t in time slot t consists of four components, namely the source-relay link state U^t, the relay-destination link state D^t, the data buffer state B^t, and the battery state E^t.

The transmission states U^t and D^t indicate the quality of the links at time t. The source-relay link state $U^t = \{u_1^t, u_2^t, \ldots, u_N^t\}$, where u_i^t is the discretized achievable transmission rate from UE_i to the relay RN at time t. Similarly, the relay-destination link state $D^t = \{d_1^t, d_2^t, \ldots, d_N^t\}$, where d_i^t is the the discretized achievable transmission rate from RN to DE_i at time t. The status of the links in U^t and D^t are independent random variables with the sample space L_{link}.

The data buffer state $B^t = \{b_1^t, b_2^t, \ldots, b_N^t\}$, where b_i^t is the amount of cached data packets for pair (UE_i, DE_i), namely, the packets that are received from UE_i and have not yet been forwarded to DE_i. Notice that at any time t, for each source-destination pairs (UE_i, DE_i), the amount of the cached data can not exceed the quota q_i allocated for the pair, i.e. $0 \leq b_i^t \leq q_i$ for $i = 1..N$ and each t.

The energy state E^t indicates the remaining energy in RN for forwarding packets to the destination nodes. At time t, the battery state E^t depends on 1) the previously remaining energy E^{t-1}, 2) the energy gain EG^{t-1} from the environment, 3) the energy cost EC^{t-1}, and 4) the battery capacity E_{max}. The energy gain submits to a certain energy harvest model, and the energy cost depends on the amount of packets that are successfully transmitted to the destination nodes. Details will be introduced in the transition function.

B. ACTION SPACE

Given a system state s, the RN decides whether to transmit packets or not, and determines which link should be selected for transmission. Formally, the action space $A = \{wait\} \cup \{rcv_i \mid i = 1..N\} \cup \{fwd_i \mid i = 1..N\}$. At any time t, the RN either waits for more harvested energy, or receives packets from a source node UE_i (rcv_i), or forwards packets to a destination node DE_i (fwd_i). Note that the above receiving and forwarding actions do not include the packet amount as a parameter. The packet amount of each action is determined by following factors: the current link state, the current buffer state and the remaining energy in the battery. The RN always receives or forwards, if it is not waiting, as many packets as possible under the constraints of the transmission rate of the links, the buffer quota and the energy. Hence, at time t, there are totally $2N + 1$ candidate actions in A. The action rcv_i^t is to receive from source node UE_i as many packets as possible at time t, under following constraints:

1) the amount of the received packets can not exceed the source-relay transmission rate u_i^t,
2) by receiving the packets, the total amount b_i^{t+1} of the buffered data in RN can not exceed the quota q_i.

Therefore, the amount of the data packets received by taking action rcv_i^t is $min(u_i^t, q_i - b_i^t)$. Similarly, the action fwd_i^t is to forward to user DE_i as many data as possible at time t, with following constraints:

1) the amount of the forwarded packets can not exceed the relay-destination transmission rate d_i^t,

2) the amount of the forwarded packets can not exceed the total amount b_i^t of the buffered packets in RN. In other words, only the data that is already in RN can be forwarded to DE_i,
3) the remaining energy E^t is enough for the transmission.

Therefore, the amount of the data packets forwarded by taking action fwd_i^t is $min(d_i^t, b_i^t, E^t)$.

C. TRANSITION FUNCTION

The transition function $Tr : S \times A \rightarrow S$ is defined as follow: for a state $s^t = (U^t, D^t, B^t, E^t)$ at time t, by taking the action a^t, the successor system state $s^{t+1} = (U^{t+1}, D^{t+1}, B^{t+1}, E^{t+1})$:

- The transmission states $U^{t+1} = \{u_1^{t+1}, u_2^{t+1}, \ldots, u_N^{t+1}\}$ and $D^{t+1} = \{d_1^{t+1}, d_2^{t+1}, \ldots, d_N^{t+1}\}$ where $u_i^{t+1}, d_i^{t+1} \in L_{link}$. The transition of the link states are decided by the environment model. In the experiment section, we will verify our approach with multiple environment models. Note that in many real applications, the environment model is unavailable to the system.
- The buffer state $B^{t+1} = \{b_1^{t+1}, b_2^{t+1}, \ldots, b_N^{t+1}\}$ where

$$b_i^{t+1} = \begin{cases} min(b_i^t + u_i^t, q_i) & a^t = rcv_i^t \\ max(b_i^t - min(d_i^t, E^t), 0) & a^t = fwd_i^t \\ b_i^t & \text{otherwise} \end{cases} \quad (1)$$

  If the action a^t is to receive data from UE_i, the amount of packets finally received by RN is determined by the current source-relay transmission rate and the remaining buffer quota for UE_i at time t. If the action a^t is sending data to DE_i, the amount of packets received by DE_i is determined by the current relay-destination transmission rate, the remaining cached data, and the remaining energy in the relay at time t.
- The energy state $E^{t+1} = min(E_{max}, E^t + EG^t - EC^t)$, where E_{max} is the maximum energy volume of the battery, EG^t is the energy gain from the environment, a random variable submitting to the energy harvest model, and EC^t is the energy consumption at time t,

$$EC^t = \begin{cases} min(d_i^t, b_i^t, E^t) & a^t = fwd_i^t \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

D. REWARD FUNCTION

The reward function $R : S \times A \rightarrow R$ is defined as follows: given a system state s^t, and and action a^t, suppose that $Tr(s^t, a^t) = s^{t+1}$, and the reward

$$R(s^t, a^t) = \begin{cases} r_u \cdot (b_i^{t+1} - b_i^t) & a = rcv_i^t, i = 1..N \\ r_d \cdot (b_i^t - b_i^{t+1}) & a = fwd_i^t, i = 1..N \\ 0 & a = wait \end{cases} \quad (3)$$

where r_u and r_d are unit rewards for successfully receiving and forwarding data, respectively.

The expected reward of a system state s^t is defined as

$$R(s^t) = \sum_{a^t \in A} Pr(a^t|s^t)\big(R(s^t, a^t) + \gamma \sum_{s^{t+1} \in S} Pr(s^t, a^t, s^{t+1})R(s^{t+1}) \quad (4)$$

where $Pr(a^t|s^t)$ is determined by the given decision policy, and $Pr(s^t, a^t, s^{t+1})$ depends on the environment model, which might be unknown in a real-world domain.

E. OPTIMIZATION PROBLEM

With the above state space, action space, transition function and reward function, we formulate our task of link selection as following optimization problem. For each system state s^t, the system needs to find an optimal policy $\pi : S \rightarrow A$ that maximizes the expected long-term reward

$$\pi^* = \arg\max_{\pi} \left[\mathbb{E} \sum_{t=0}^{\infty} \gamma^t R(s^t, a^t|\pi)\right] \quad (5)$$

For any given state s^t, the policy should give an optimal action,

$$\pi^*(s^t) = \arg\max_{a^t \in A} \left[R(s^t, a^t) + \sum_{k=1}^{\infty} \gamma^k R(s^{t+k}, \pi^*(s^{t+k}))\right] \quad (6)$$

where $s^{k+t} = Tr(s^{t+k-1}, \pi^*(s^{t+k-1}))$.

Such an optimization problem certainly can be solved by some traditional methods, such as value iteration and policy iteration. However, as mentioned in the previous section, these traditional methods require the explicit transition model for optimization. But in many environments, the transition model is unavailable to the relaying system. For instance, the transmission quality of a link at certain time is affected by many complex factors, which are difficult, sometimes even impossible to be modeled. A straightforward solution is to estimate the transition model by sampling, but in this way the performance becomes very sensitive to the sampling quality. In particular, for a dynamic environment where the transition model might change over periods, it is difficult for the sampling-based estimation to achieve satisfying precision. Moreover, the state spaces in real-world domains are usually large, which results that these model-based methods suffer from the computational complexity. Therefore, in this paper we deploy the Deep Reinforcement Learning (DRL) and approximate the optimal policy with a deep Q-network, which requires neither explicit transition model, nor a large storage for maintaining the state-action value table.

As a well known traditional reinforcement learning method, Q-learning maintains a lookup table of the state-action values, i.e. Q-values $Q(s, a)$. In order to learn the optimal Q-value function, the Q-learning algorithm makes use of the fixed point of the Bellman equation for the Q-value function,

$$Q^*(s, a) = \sum_{s' \in S} T(s, a, s')(R(s, a, s') + \gamma \max_{a' \in A} Q^*(s', a')) \quad (7)$$

The basic Q-learning looks up in the Q-table to decide an optimal action a^* to take (the one with highest Q-value $Q^*(s, a)$), given a system state s. However, for a domain with a large state space, the computation of the Q-table is quite consuming both in time and space. DRL uses a deep neural network with weights θ as a function approximator, referred as a Q-network. A Q-network can be trained by minimizing a sequence of loss functions $L_i(\theta_i)$ that changes at each iteration i,

$$L_i(\theta_i) = \mathbb{E}_{s,a\sim\rho(\cdot)}\left[(y_i - Q(s, a; \theta_i))^2\right] \tag{8}$$

where $y_i = \mathbb{E}_{s'\sim\mathcal{E}}\left[r + \gamma \max_{a'} Q(s', a'; \theta_{i-1})|s, a\right]$ is the target for iteration i and $\rho(s, a)$ is a probability distribution over states and actions. Note that above r is the instant reward, which corresponds to $R(s, a)$ in our formulation. The distribution $\rho(s, a)$ is selected by an ε-greedy strategy. In each training iteration, the system selects a random action with probability ε, otherwise, follows the greedy strategy and selects the action with highest estimated Q-value. To alleviate the problems of correlated data and non-stationary distributions, DQN maintains an experience replay memory D. The system executes the selected action and observe the reward and successor state, and then stores the transition in D. Then DQN performs a gradient descent step on $(y_i - Q(s, a; \theta_i))^2$:

$$\nabla_{\theta_i} L_i(\theta_i) = \mathbb{E}_{s,a\sim\rho(\cdot),s'\sim\mathcal{E}}\Big[\big(r + \gamma \max_{a'} Q(s', a'; \theta_{i-1}) - Q(s, a; \theta_i)\big)\nabla_{\theta_i} Q(s, a; \theta_i)\Big] \tag{9}$$

with a random batch of previous transitions.

IV. PROPOSED DRL-BASED ACCESS CONTROL WITH PRETRAINING

In this section, we present the DRL-based access control for a buffer-aided relaying system with energy harvesting. The architecture of the system is shown in Figure 2. As shown in Algorithm 1, our method aims to maximize the long-term system reward

$$G^t = \sum_{k=0}^{\infty} \gamma^k R(s^{k+t}, a^{k+t}),$$

with the discount factor $\gamma \in (0, 1)$.

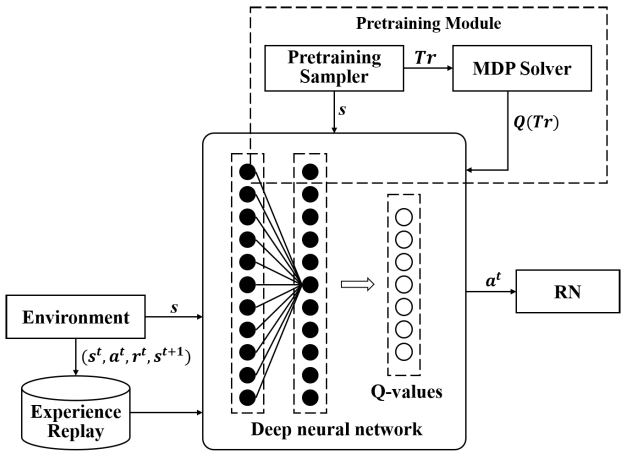


FIGURE 2. DRL-based access control with pretraining.

Algorithm 1 DRL-Based Access Control With Pretraining

Initialization
Initialize the cache quotas and the decay rate
Initialize the Q-network to random weights
Pretraining
Sample state transition model P
for *episode* $= 1$ to M_p **do**
  Initialize system state s^0
  for $t = 1$ to T_p **do**
    With probability ε set a^t to a random action, otherwise set $a^t = \max_a Q(s^t, a^t; \theta)$
    Take action a^t, get reward r^t and next state s^{t+1} according to transition model P
    Store transition (s^t, a^t, r^t, s^{t+1}) in D
    Sample random minibatch of transitions from D
    Perform gradient descent and update the network
  end for
end for
return the parameter θ_P of current network.
Training
Initialize the parameter $\theta := \theta_P$ and $D := \emptyset$
for *episode* $= 1$ to M **do**
  Initialize system state s^0
  for $t = 1$ to T **do**
    Perform ε-greedy step
    Perform gradient descent with D, and fine tune θ on the last layer
    Perform gradient descent and fine tuning on Q
  end for
end for

A. STRUCTURED FEATURES OF SYSTEM

To learn an effective access control policy for the system, the DQN should be trained first. Usually, the weights θ of the DQN is randomly initialized before training, which means the Q-values $Q(s, a)$ is randomly initialized. Then by training using data sampled from real system, effective weights θ of the DQN can be found. However, with this initializing method, it may take a long training time to find an effective DQN weights θ. Actually, for the considered system, there are some structured features which are independent of the environment model (the channel gain processes and energy harvesting models). For instance, it is always preferred to select a relay-destination link if the power (data) in the battery (date buffer) is full or almost full no matter what transition function $Tr : S \times A \rightarrow S$ is for the system. When the power (data) in the battery (date buffer) is empty or almost empty, it is a better choice to receive data packets.

These features indicates that when s is a state in which the power (data) in the battery (date buffer) is full or almost full, the Q-values $Q(s, a)$ with action a choosing to transmit packets from the RN to a certain destination should be assigned a larger value. While when s is a state in which the power (data) in the battery (date buffer) is empty or almost empty, the Q-values $Q(s, a)$ with action a choosing to receive packets from a certain source should be assigned a larger value. Note that these preferences hold for any system transition function $Tr : S \times A \rightarrow S$. With these observations, we proposed a pretraining method to assign a proper initial Q-values $Q(s, a)$ for the DQN.

B. PRETRAINING WITH HYPOTHETICAL ENVIRONMENT

The pretraining module consists of a sampler and a MDP solver. The pretraining sampler first samples a transition model P, and then samples transitions one by one according to P. This pretrained model is used to initialize and fine tune the network parameters in order to leverage faster training improvement. The ε-greedy step and gradient descent step are similar with the original deep Q-learning. The transition model in the pretraining process is a hypothetical one, which is sampled randomly in the experiment. The ε-greedy step and gradient descent step are similar with the original deep Q-learning. With probability ε the system takes arbitrary action to explore in the hypothesis environment, otherwise, the system will take the action with the maximum state-action value (estimated by the Q-network). The environment gives feedback to the system, including the successor state and direct reward. To weaken the correlation among the training samples, the transitions during the process will be restored in an experience replay set, from which the following training transitions are sampled for the Q-network update. The update step is to perform a gradient descent on $(y_j - Q(\phi_j, a_j; \theta))^2$ with respect to the network parameters θ. When the average performance is satisfying enough or maximum training time is reached, we fix the network and return the pretrained parameter.

The model might be quite different with the one in the authentic environment, yet the core ability of the decision making under the framework can be captured by the Q-network. It turns out that with the pretrained Q-network, only a fine tuning is needed to enable good performance in the authentic environment. Note that in the pretraining stage, the Q-table can be calculated or estimated by any given MDP solver. In our experiment, we just deploy the deep neural network to achieve the approximation.

The pretraining mechanism accelerates the learning process, and guarantees the efficiency even in a dynamic environment where the transition model changes over time. For instance, consider an environment where the quality of the links submit to a dynamic distribution P, which changes every M time slots. The tasks with different P share some common structural features of the system, but without further training or fine tuning, they are after all different tasks for the neural network. It is verified by our experiment that the deep neural network does not perform well if it is directly deployed in a brand new environment. So in this case, increasing training episodes does not help improve the performance and the robustness. In contrast, the fine tuning of the pretrained model eases the pain of context switching. Within each cycle of the dynamic transition model, we tune the pretrained model for the first m time slots, and the neural network can achieve good performance under the new distribution for the rest $M - m$ time slots. Our experiment result shows that the pretraining mechanism can dramatically improve the learning efficiency and the performance stability.

C. FINE TUNING WITHIN AUTHENTIC ENVIRONMENTS

Once the model is well pretrained with a hypothetical environment, it is ready for further training in authentic environments. As shown in Algorithm 1, during the training process, the pretrained model only needs to be fine tuned. Although the transition model in authentic environment may be quite different from the hypothetical environment, they share some common structural features of the relaying system. By pretraining in the hypothetical environment, the deep neural network well captures the structural features of the relaying system, and is able to quickly learn how to maximize the expected long-term reward even in a different environment. Verified by the experiments on different environment models, the training efficiency can be dramatically improved by the pretraining mechanism.

Note that to avoid overfitting, the experience replay of the deep model needs to be reset before further training in an authentic environment, as in Line 15 of Algorithm 1.

D. EXTENSION: STRATEGY FOR POWER OUTAGE

In the experiment to introduce in following sections, the relaying system sometimes needs to work in an environment short of energy. The algorithm above for sure can deal with this situation by learning, resulting that the system decides to wait for energy harvest when the power is used up. Nevertheless, some other strategy can be considered to deal with the specific situation of power outage. In this section, we extend the above main algorithm by embedding power outage prevention into the reward function.

The reward function (3) is revised to

$$R(s^t, a^t) = R_{tr}(s^t, a^t) + R_{pw}(s^t, a^t) \tag{10}$$

where

$$R_{tr}(s^t, a^t) = \begin{cases} r_u \cdot (b_i^{t+1} - b_i^t) & a = rcv_i^t, i = 1..N \\ r_d \cdot (b_i^t - b_i^{t+1}) & a = fwd_i^t, i = 1..N \\ 0 & a = wait \end{cases} \tag{11}$$

and

$$R_{pw}(s^t, a^t) = r_p \cdot \delta(E^{t+1}), \tag{12}$$

the r_p is the unit reward for energy sufficiency, and δ is the reward model for preventing the power outage. In our experiment, we use a sigmoid function as our power reward model.

We compares the above algorithm with and without the energy strategy and give the experiment details in Section V.

Note that beside encoding the power sufficiency policy into the reward function, there are other ways to implement given policies in a reinforcement learning based framework, and we will leave them for future work.

V. EXPERIMENTS

In this section, we demonstrate the performance of the algorithm with a series of experiments. For simplicity, we consider an instance of the system model above with the number of communication pairs $N = 2$. Note that the experiments carried out in this section can be directly applied to the cases with $N > 2$ communication pairs. Since in practical wireless communication systems, even if the channel state information is perfectly known at the transmitter, only several discrete communication rate levels are supported [36] which corresponding to different channel coding rates and modulation orders, e.g., for LTE, only several turbo (LDPC) code rates and modulation schemes (BPSK, QPSK, 16QAM, etc.) are adopted. Thus, in this paper, we also adopt a discrete transmission rate model $L_{link} = \{0, 1, 2, 3\}$. Here, $L_{link} = 0$ can be understood as that the channel condition is too bad, even if transmission with the lowest code rate and modulation order, a data packet cannot be successfully delivered. $L_{link} = 1$ (2 or 3) means that a proper code rate and modulation order can be adopted to successfully deliver a data packet. Note that for different code rates and modulation orders, a data packet contains different amount of information. For energy harvesting, we adopted a Poisson process, which is also adopted in some other papers on energy harvesting (e.g., [5] and [6]) to model the energy arriving at the relay node. We set the arrival density as $\lambda = 1.6$. For this setup, the probability to harvest more than 3 units of energy becomes very small. Thus, we use a truncated energy harvesting model by simply assume that the energy harvest in each time slot is given by $L_{power} = \{0, 1, 2, 3\}$. In addition, the maximum energy capacity is set as $E_{max} = 10$ and the maximum cache quota for each pair $b_i = 10$ for $i = 1..N$ to ensure that the packet delay at the relay will not to too large.

The above instance seems to be a pretty small domain, yet the state space is already quite large:

$$|S| = |L_{link}|^{2N} \cdot |L_{power}| \cdot b_i^{2N} = 10,240,000,$$

which makes it impossible to efficiently maintain and update on a complete state-action value table, as in some traditional MDP solution such as policy iteration or value iteration. So we use deep neural network to estimate the state-action value function and make decisions. In the following, we present the experiments in two parts. In the first part, we compares our model with some traditional algorithms, and give the canonical variate analysis. In the second part, we discuss the improvement on learning efficiency of our pretrained model and demonstrate the generality of the acceleration on training.

A. PERFORMANCE COMPARISON

Deep neural networks well trained by good algorithm are capable to make good decision, even in domains with quite large state spaces. In the experiment, we compare the performance of our pretrained model with three baseline algorithms, namely round-robin, round-robin on pairs, and greedy approach, which are widely used in in realistic scheduling protocols and applications. For buffer-aided relay, actually it has been shown in [2] that greedy approach can usually achieve the highest throughput. However, for EH-based buffer-aided relaying, few works have been done and there is no commonly accepted access strategy that can achieve a good performance. Although the access control problem has already been investigated for some EH-based wireless communication systems, many of them also adopted the RL method to find an effective strategy. Thus, in this paper, to show the effectiveness of DRL in designing access control strategy for EH buffer-aided relaying systems, we just compared the proposed model with Round Robin and Greedy approaches.

1) **Round-robin on links.** The first baseline Round-robin on Links is to serves the users in a fixed order. The $2N$ end users take turns to be served by the relay. In other words, the decision sequence is following fixed one:

$$\{rcv_1^1, rcv_2^2, \ldots, rcv_N^N, \\ fwd_1^{N+1}, fwd_2^{N+1}, \ldots, fwd_N^{2N}, \\ rcv_1^{2N+1}, rcv_2^{2N+2}, \ldots, rcv_N^{3N}, \ldots\}.$$

2) **Round-robin on pairs.** The second baseline Round-robin on Pairs rotates the service on pairs of communication, whose decision sequence is also a fixed one:

$$\{rcv_1^1, fwd_1^2, \ldots rcv_N^{2N-1}, fwd_N^{2N}, \\ rcv_1^{2N+1}, fwd_1^{2N+2}, \ldots, rcv_N^{4N-1}, fwd_N^{4N}, \\ rcv_1^{4N+1}, fwd_1^{4N+2}, \ldots, rcv_N^{6N-1}, fwd_N^{6N}, \\ \ldots\}.$$

3) **Greedy.** The third baseline Greedy is to select the link with the best transmission rate. Suppose that $p_{max} = max(U^t \cup D^t)$, then

$$a^t = \begin{cases} fwd_i^t & p_{max} = d_i^t \\ rcv_i^t & p_{max} = u_i^t \end{cases}$$

for any time t. For links with the same transmission rate, destination-relay links have the priority to be selected.

Please note that some other heuristics can be embedded in Round and Robin or Greedy, but the basic ideas are the same. To our knowledge, except above baselines, there is not yet a commonly accepted strategy for the access control of relays with energy harvest.

First, We test our method in the six environment models, namely Env_1 to Env_6, as shown in Table 1. As in many other works such as [34], [35], the energy harvest of the RN is

TABLE 1. Different environment models.

	Transmission state model	Energy harvest model
Env_1	Uniform($a = 0, b = 1$)	Poisson($\lambda = 1.6$)
Env_2	Binomial($n = 4, p = 0.5$)	Poisson($\lambda = 1.6$)
Env_3	Gaussian($\mu = 2, \sigma = 1$)	Poisson($\lambda = 1.6$)
Env_4	Gaussian($\mu = 4, \sigma = 1$)	Poisson($\lambda = 1.6$)
Env_5	Poisson($\lambda = 0.8$)	Poisson($\lambda = 1.6$)
Env_6	Poisson($\lambda = 2$)	Poisson($\lambda = 1.6$)

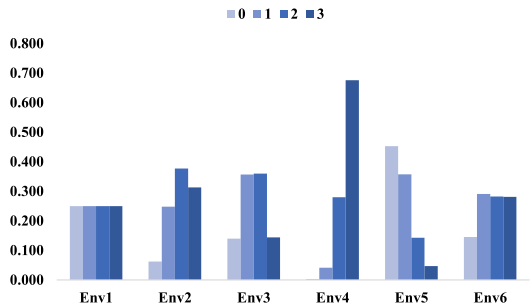


FIGURE 3. Different transition models of transmission rates.

modeled as a Poisson process. The transmission rates of the links are independent random variables that submit to the given environment model. The distribution of the link states is shown in Figure 3.

Second, we compare the performance of the algorithm with traditional methods in terms of average reward, in each environment model. As shown in Figure 4, in all the environment models, our method dominates others and shows the best performance. For instance in Figure 4(a), the link transmission rate of each link distribute on the sample space uniformly. The average reward of our model is significantly higher than the other three traditional methods. In Figure 4(c) our method is slightly better than greedy and much better than two Round and Robin baselines. For Env_4 in Figure 4(d), the greedy method shows the worst performance, even worse than round and robin. One of the reasons is that the transmission rate are usually quite high in Env_4, and for a pair with good communication, the buffer quota can be easily used up, and the greedy algorithm still tries to select the pair and produces redundant actions.

Third, we compare the competing algorithms on the average packet delay in Figure 5. The metric average packet delay $pd(T)$ for a time period T is defined as

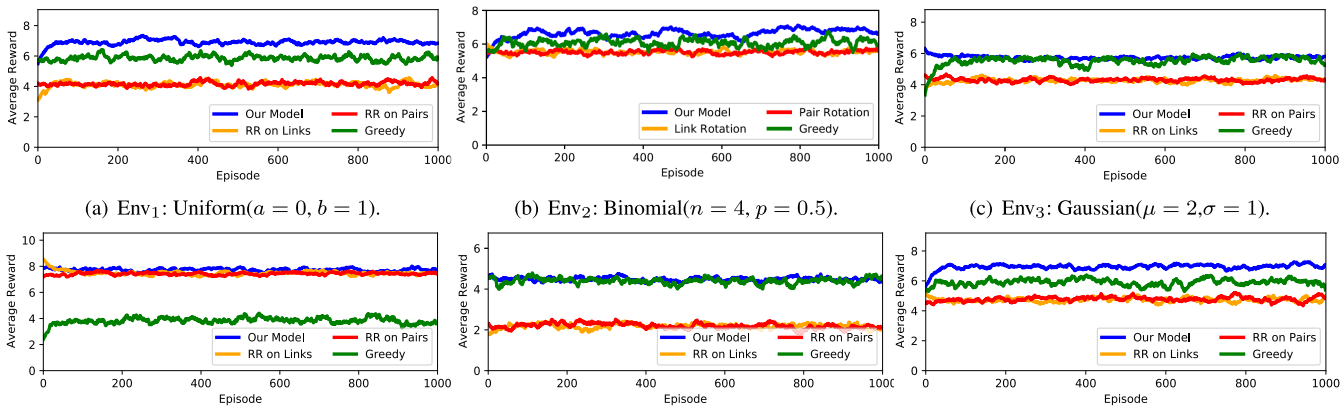
$$pd(T) = \frac{num_{fail} + \sum_{t=1}^{T} \sum_{i=1}^{N} b_i^t}{T}$$

where num_{fail} is the total number of transmission failures. As shown in the result, our method outperforms than others in almost all the environment models.

To sum up, our approach shows better performance than the three baselines for all the above environment models, in terms of the average reward and the average packet delay.

B. CANONICAL VARIATE ANALYSIS ON EH MODEL

The energy harvest of the RN is modeled as a Poisson process as in other works such as [35]. The harvested power is a random variable under a given Poisson distribution. It is worth noting that the energy harvested at the current time can only be used for subsequent time slots, that is, the harvest-store-use (HSU) mode is used in this paper. We present the canonical variate analysis of our algorithm, discussing the performance of the algorithm with different energy harvest rates. Under the premise that the source-relay transmission state and relay-destination transmission state are sampled from L_{link} according to a a common distribution, and the maximum energy capacity $E_{max} = 10$, and the maximum cache quota for each pair $b_i = 10$ for $i = 1..N$. We consider $Poisson(\lambda = 0.8)$ which indicates that the harvested power is sufficient, and $Poisson(\lambda = 2.8)$ where the harvested power is relatively insufficient. As shown in Figure 6, our method outperforms all the baselines in both EH models. The improvement is more obvious when the harvested power is insufficient.



(a) Env_1: Uniform($a = 0, b = 1$).
(b) Env_2: Binomial($n = 4, p = 0.5$).
(c) Env_3: Gaussian($\mu = 2, \sigma = 1$).
(d) Env_4: Gaussian($\mu = 4, \mu = 1$).
(e) Env_5: Poisson($\lambda = 0.8$).
(f) Env_6: Poisson($\lambda = 2$).

FIGURE 4. Performance comparison in Env_1 - Env_6: average reward.

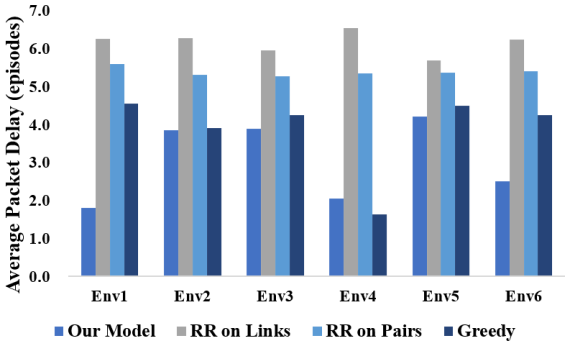
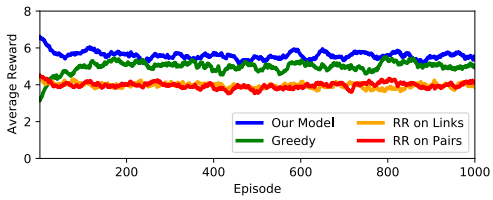
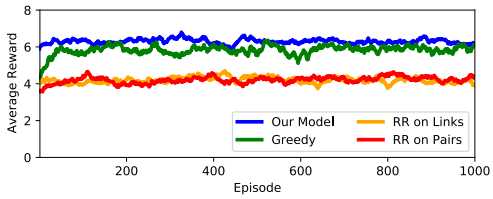


FIGURE 5. Performance comparison in Env_1 - Env_6: average packet delay.

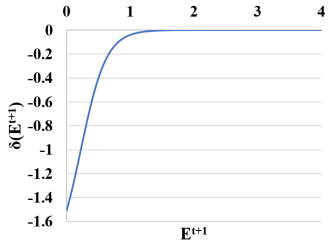


(a) EH: Poisson($\lambda = 0.8$).

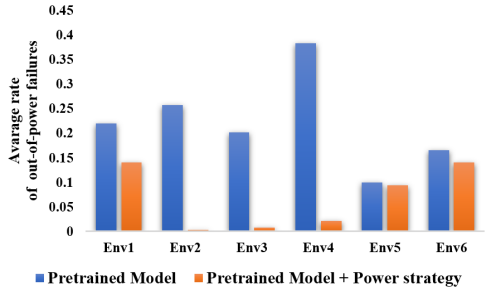


(b) EH: Poisson($\lambda = 2.8$).

FIGURE 6. Performance comparison with different energy harvest models.



(a) Power shortage penalty



(b) Average out-of-power failure rate

FIGURE 7. Strategy for power outage prevention.

TABLE 2. Average number of training episodes.

	Baseline	Pretrained model	Improvement
Env_1	609	75	87.68%
Env_2	390	102	73.85%
Env_3	315	180	42.86%
Env_4	244	156	36.07%
Env_5	233	210	9.87%
Env_6	221	89	59.73%

C. POWER OUTAGE PREVENTION

As stated in previous sections, when the power in the battery is used up, the relaying system has to suspend and wait until enough energy is harvested, which is not good enough for data transmission. We embed a policy for preventing the power outage into the reward function and make the access control smarter. The new reward function is,

$$R(s^t, a^t) = R_{tr}(s^t, a^t) + R_{pw}(s^t, a^t) \tag{13}$$

where $R_{tr}(s^t, a^t)$ is the same as the previous $R(s^t, a^t)$ in above groups of experiments, and

$$R_{pw}(s^t, a^t) = r_p \cdot \delta(E^{t+1}), \tag{14}$$

the r_p is the unit reward for energy sufficiency, and δ is the reward model for preventing the power outage.

Note that there is no existing work on how to embed a given policy of power outage prevention into the reward function for a deep reinforcement learning based relaying system with harvest energy. In our experiment, we set the $r_p = r_d$ and

$$\delta(E^{t+1}) = \frac{1}{0.5 + 0.125 \cdot e^{-5(E^{t+1}-0.5)}} - 2$$

Part of the curve of $\delta(E^{t+1})$ is shown in Figure 7(a). When the power left in $t+1$ is going to be close to 0, the penalty of around $1.5 * u_d$. In other words, only if more than 1.5 packets are going to successfully forwarded, the relay will decide to forward even the power is almost used up. Otherwise, the relay just waits for more energy harvest. If the power left in $t+1$ is going to be larger than $1.2 * u_d$, then the penalty is almost 0.

The performance comparison is shown in Figure 8. The result indicates that the average reward of our model with strategy is a bit lower than the algorithm without the strategy, due to the extra waiting when the battery is short of power, and better then the baselines in almost all environments. But as shown in Figure 7(a), the rate of transmission failure due to power outage is dramatically reduced.

D. ACCELERATION BY PRETRAINING

As stated previously, we first sample a transition model for pretraining, and test the performance in an authentic environment. The comparison of our algorithm with original deep Q-learning is shown in Figure 9. The deep model is pretrained in a hypothetical environment for 500 iterations, and then compared with a deep model without pretraining for 1000 training iterations. The learning efficiency is dramatically improved by the pretraining.

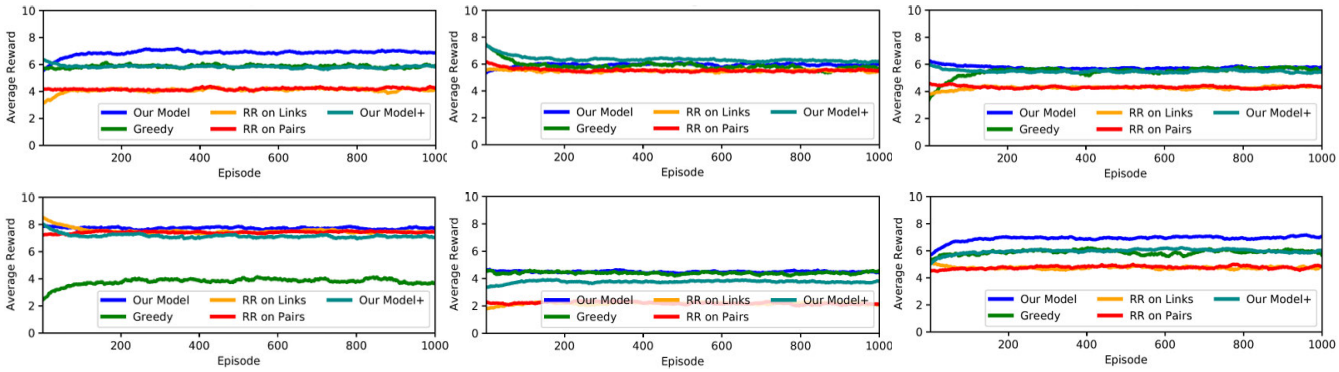
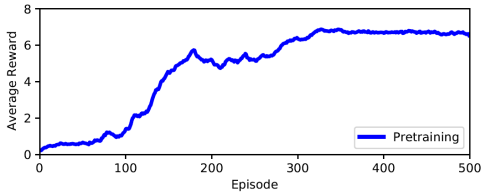
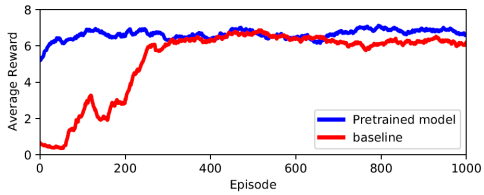


FIGURE 8. Performance of our model with power saving strategy (*our model+*).



(a) Pretraining curve



(b) Learning efficiency comparison

FIGURE 9. Deep models with and without pretraining.

To demonstrate the generality of the pretraining mechanism, we compare in Figure 10 our algorithm with the unpretrained deep models within different authentic environments. The result verifies the generality of the acceleration - no matter how the authentic environment varies, the pretained model always converges much faster than the baseline. In Table 2, we give the average training episodes in of the two methods in Env_1 to Env_6. For this MDP problem with a relatively large state space, the neuron network takes hundreds of episodes to be well trained, without pretraining. But the pretraining successfully eases the pain of learning complexity.

VI. CONCLUSION AND FUTURE WORK

In this paper, we formulate the link selection problem in the EH based buffer-aided relaying communication systems as a RL problem to maximize the long-term average system utility. The DQN is adopted to find an optimal access control strategy. Experiment results showed that the transmission policy obtained by using our proposed model can achieve better performance compared with the traditional methods, and is general enough to be used in different environments, which demonstrate the adaptive ability of our proposed model.To reduce the training time, an effective pretraining algorithm is proposed to accelerate the convergence of the proposed Deep Q-learning method. It is shown that the proposed pretraining method can significantly reduce the training time. Thus, it was proved that the considered system have some general structural features and adaptive training method can be adopted in practical time varying systems whose transition model changes with time.

Moreover, in this paper, we embed some domain knowledge in terms of a power policy into the reward function of the deep learning. There are also some other ways to embed explicit policies or domain knowledge into the learning frameworks, and we will leave them for future works.

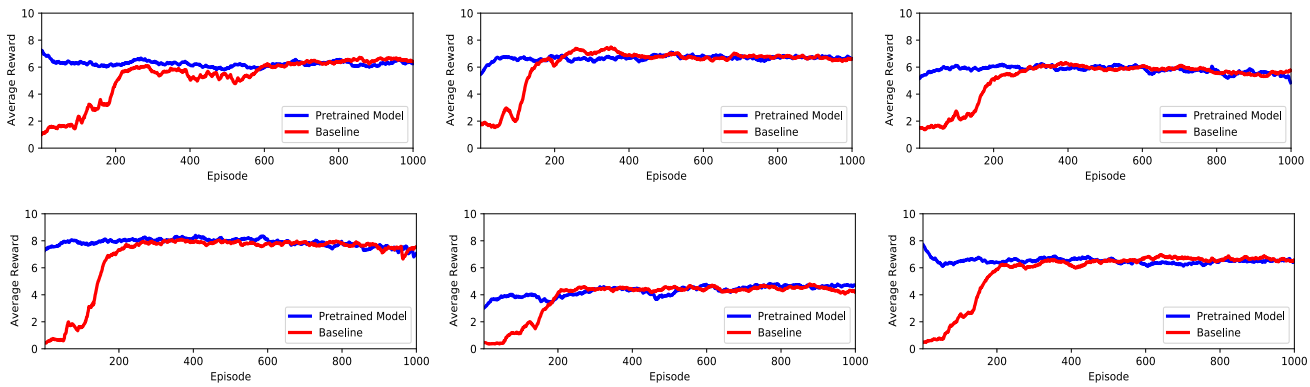


FIGURE 10. Learning efficiency comparison in Env_1 - Env_6.

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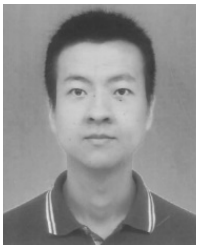

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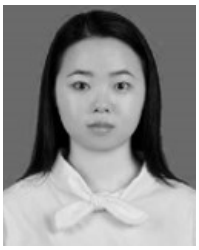

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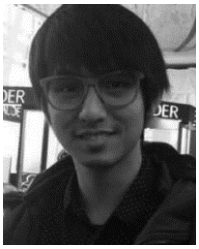

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